Core practical – Electrolysis of copper sulfate solution

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Practical Objective

To electrolyse copper sulfate solution using inert (graphite) electrodes and copper electrodes.

Apparatus

- eye protection
- emery paper
- low voltage d.c. supply
- ammeter
- variable resistor
- connecting leads
- crocodile clips
- 100 cm^3 beaker
- stop clock
- two graphite rods
- two pieces of copper foil
- copper sulfate solution
- distilled water
- access to propanone (in a fume cupboard)
- access to a balance

Safety

- Wear eye protection.
- Propanone is an irritant. It is highly flammable; there must not be any naked flames in the laboratory**.**

Your teacher may watch to see if you can:

- carefully control variables during investigations
- make accurate measurements.

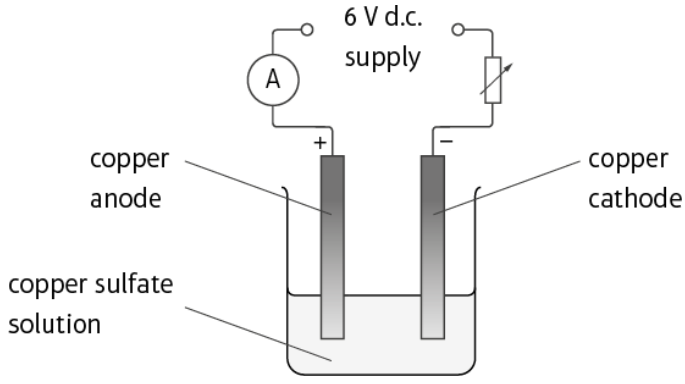
Method 1 – Using copper electrodes

A Select two pieces of copper foil to use as electrodes and clean them with emery paper. Label one of the electrodes as ‘anode’ and the other as ‘cathode’.

B Measure and record the mass of each electrode.

C Half fill the beaker with copper sulfate solution.

D Set up the circuit as shown in the diagram.



E Turn the power on and adjust the variable resistor to give a current of 0.2 A. Record the current. Leave the power on for 20 minutes, adjusting the variable resistor to keep the current constant, if necessary.

F Then turn off the power and remove the electrodes from the beaker.

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G Gently rinse the electrodes with distilled water and then dip them into propanone. Remove the electrodes from the propanone and gently shake them until the propanone evaporates.

H Measure and record the masses of the dry electrodes.

I Repeat the experiment using currents of 0.3 A, 0.4 A and 0.5 A.

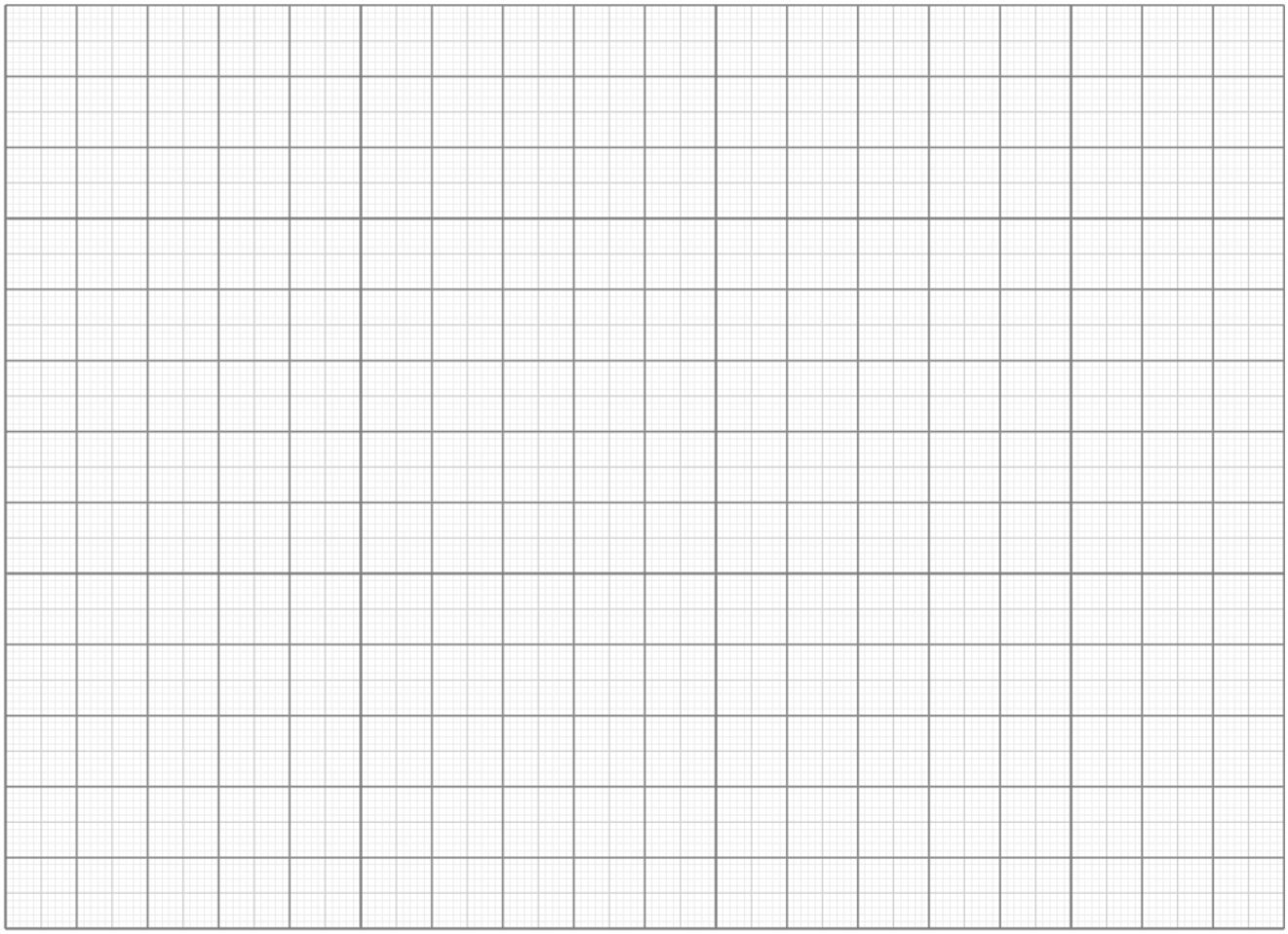
Recording your results

1 Record your results in the table below, including the change in mass of each electrode

Current (A)	Mass of anode at start (g)	Mass of anode at end (g)	Change in mass of anode (g)	Mass of cathode at start (g)	Mass of cathode at end (g)	Change in mass of cathode (g)

Considering your results/conclusions

2 Plot a graph of change in mass of the anode against the current. Draw a line of best fit through the points. Plot a second graph on the same axes showing the change in mass of the cathode against the current. Again, draw a line of best fit through these points.



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3 Describe the relationship between the change in mass at each electrode and the current.

4 Explain the changes in mass of each electrode.

5 Use your graph to predict the change in mass of the anode when the current is 0.35 A.

6 Suggest a reason why the change in mass at the cathode is not the same as the change in mass at the anode when the same current is used.

7 Describe how you could improve the experiment to be more certain that the data collected is correct and free from error.

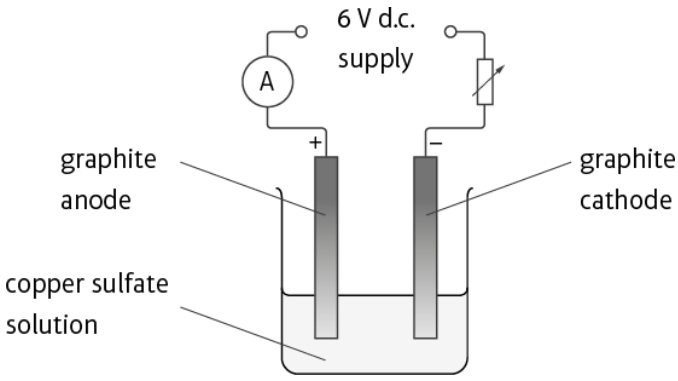
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Method 2 – Using inert electrodes

A The diagram on the right shows the circuit used for the electrolysis of copper sulfate solution with graphite electrodes. Set up the circuit as shown.

B Turn the power on.

C Observe what happens at each electrode.



Recording your results

1 Record your observations and the name of the product formed at each electrode.

Considering your results

2 Explain the formation of the product at each electrode.

H **3** Write a half equation for the formation of the product at each electrode and classify each reaction as oxidation or reduction.

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